

Detecting AI-Generated Voice Phishing under Telephony Degradation: A Robustness Study of Spectrogram-Based Deepfake Audio Detection

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Abstract—Accessible voice-cloning tools have driven a rise in voice-phishing (“vishing”) fraud, yet deepfake-audio detectors are typically trained and evaluated on clean, high-quality speech — while real fraudulent calls arrive over noisy, compressed, narrowband telephone channels. This paper studies detector robustness in that neglected regime. Using the ASVspoof 2019 Logical Access dataset, we define a set of telephony degradations — additive noise, narrowband simulation, μ -law codec quantisation, and a combined phone-call condition — and use them both to measure robustness and, as training-time augmentation, to improve it. We compare a lightweight convolutional baseline with a pretrained Audio Spectrogram Transformer (AST). A clean-trained CNN achieves 3.50% Equal Error Rate on clean audio but collapses to near-chance (50%) under noise and phone-channel conditions; augmentation restores it to 25.7% on the phone-call condition, but at a cost to clean accuracy. The pretrained AST is far more robust and, uniquely, augmentation-friendly: it reaches 8.58% on the phone-call condition — an 83% relative reduction over the clean-trained CNN — while improving clean-audio accuracy to 1.33%. The results indicate that robustness is bounded more by model capacity and pretraining than by the augmentation strategy alone.

I. INTRODUCTION

Recent advances in generative speech synthesis have made voice cloning cheap, fast, and convincing: a few seconds of a target’s speech is now enough to produce fluent, natural-sounding audio in that person’s voice. This capability has been rapidly weaponised for voice phishing (“vishing”), in which an attacker impersonates a trusted individual — a relative, a manager, a bank — over a phone call to commit fraud, and reported losses from such AI-enabled impersonation scams have grown accordingly. [9] A substantial body of work now exists on detecting synthetic speech, and public benchmarks such as ASVspoof have driven rapid progress. However, these detectors are overwhelmingly trained and evaluated on clean, high-quality audio. Real vishing calls do not arrive this way: they pass through lossy codecs, narrowband telephone channels, and noisy environments, all of which distort the very acoustic cues a detector relies on. The conditions under which detection matters most are therefore precisely those in which detectors are least tested.

This work investigates that gap through three research questions.

- RQ1: how much does telephony-style degradation harm a detector trained only on clean audio?
- RQ2: can training-time augmentation with the same degradations restore robustness?
- RQ3: does a higher-capacity, pretrained model change the picture?

To answer these, we make three contributions:

- We construct a reproducible telephony-degradation benchmark over the ASVspoof 2019 Logical Access dataset, comprising additive noise, narrowband channel simulation, codec-style quantisation, and a combined “phone-call” condition.
- We demonstrate that a clean-trained detector collapses to near-chance performance under these conditions, while degradation-based augmentation restores usable performance at the cost of reduced clean-audio accuracy.
- We show that a pretrained Audio Spectrogram Transformer improves robustness and eliminates the clean-versus-robustness trade-off observed in the small convolutional baseline.

Beyond these specific findings, the work is motivated by a practical concern: detection systems that perform well in the laboratory but fail on real phone audio offer little protection to the people most exposed to voice-phishing fraud. Establishing how such systems behave under realistic channel degradation is therefore a prerequisite for deploying them where they are actually needed.

II. RELATED WORK

A. Synthetic-speech detection and benchmarks

The detection of synthetic and manipulated speech has been driven largely by the ASVspoof initiative, which provides standardised datasets and protocols for anti-spoofing research. The 2019 edition [1] separates logical access (LA) attacks, generated by text-to-speech and voice-conversion systems, from physical access (replay) attacks. Its LA subset is now a common benchmark for deepfake-audio detection: crucially,

the evaluation set contains attack types not present during training, so systems are assessed partly on their ability to generalise to unseen synthesis methods. Equal Error Rate (EER) is the standard metric. This work adopts ASVspoof 2019 LA as its dataset and EER as its primary measure, but departs from the standard protocol by evaluating detectors under deliberately degraded acoustic conditions rather than the clean audio on which the benchmark is normally scored.

B. Detector architectures

A range of architectures has been proposed for this task. Early systems applied convolutional neural networks to spectrogram or cepstral representations, treating detection as an image-classification problem. Later work moved to raw-waveform models: RawNet2 [4] learns directly from the waveform using fixed sinc-shaped filters as its first layer, removing the need for hand-designed features, and AASIST [3] extends this direction with spectro-temporal graph attention, achieving strong results on ASVspoof 2019 LA. More recently, self-supervised front-ends such as wav2vec 2.0 [5], pretrained on large quantities of unlabelled speech, have been used as feature extractors and paired with anti-spoofing back-ends to substantial effect. In parallel, the Audio Spectrogram Transformer (AST) [2] applies a purely attention-based architecture to spectrograms and, by transferring weights pretrained on large audio corpora, reaches state-of-the-art performance on general audio classification. The present work uses a lightweight spectrogram CNN as a baseline and AST as a high-capacity, pretrained comparator, allowing the effect of model capacity and pretraining on robustness to be examined directly.

C. Robustness and data augmentation

A recurring limitation of the above systems is that they are typically trained and evaluated on clean, high-quality audio, whereas deployment conditions involve codec compression, channel transmission, and background noise. Data augmentation is the standard remedy. RawBoost [6] augments raw audio with convolutive, impulsive, and stationary noise specifically to emulate the device, channel, and codec artefacts encountered in telephony, without requiring any external data. Augmentation techniques from adjacent fields, such as SpecAugment, are also widely used. This work is aligned with that line of research but narrower and more controlled: rather than a general-purpose augmentation scheme, it defines an explicit set of telephony degradations, uses them both to measure robustness as distinct evaluation conditions and to improve it through augmentation, and quantifies the gap between the two.

D. Generalisation and real-world audio

Finally, several studies question how well detectors trained on curated benchmarks transfer to real conditions. Müller et al. [7] introduce the “In-the-Wild” dataset of genuine and deepfake speech collected from public sources and show that detectors trained on ASVspoof generalise poorly to it, with large performance drops. More broadly, large-scale pretrained

audio models such as PANNs [8], trained on diverse real-world audio, provide transferable representations that tend to be more robust than models trained from scratch on a single clean corpus — context that helps explain the behaviour of the pretrained model examined here. This work does not yet evaluate on in-the-wild data, but the observed robustness of a pretrained transformer is consistent with these findings, and such evaluation is identified as future work.

III. METHODOLOGY

A. Dataset

All experiments use the ASVspoof 2019 Logical Access (LA) dataset [1], which pairs genuine (bonafide) speech drawn from the VCTK corpus with spoofed speech produced by a range of text-to-speech and voice-conversion systems, distributed as 16 kHz FLAC files. The training partition contains 25,380 utterances, of which 2,580 are bonafide and 22,800 spoofed — a class imbalance of roughly 9:1 that must be handled explicitly during training. The development partition is used for validation and model selection. The evaluation partition contains attack types (A07–A19) that do not appear in training, and is reserved for assessing generalisation to unseen synthesis methods.

B. Pre-processing and features

Every utterance is fixed to a duration of 4 seconds (64,000 samples): longer clips are cropped to their first 4 seconds and shorter clips are padded with trailing silence. A random-crop strategy was considered as a mild form of augmentation but the simpler head-crop was adopted for reproducibility; this choice is revisited in Section 5. For the convolutional baseline, waveforms are converted to log-mel spectrograms with 80 mel bands, an FFT size of 512, and a hop length of 160 samples (10 ms at 16 kHz), followed by conversion to decibels and per-sample standardisation. The spectrogram transform is implemented as the first layer of the model, so that all degradations operate on the raw waveform before feature extraction.

C. Telephony degradations

The central experimental instrument of this work is a set of waveform-level degradations designed to emulate the telephone channel:

- **add_noise**: additive Gaussian noise scaled to a target signal-to-noise ratio (SNR), evaluated at 15 dB and 5 dB;
- **telephony_8k**: narrowband channel simulation by down-sampling from 16 kHz to 8 kHz and back, discarding spectral content above 4 kHz;
- **mulaw_codec**: μ -law companding followed by 256-level quantisation and expansion, approximating the lossy quantisation of G.711-style telephony codecs;
- **phone_call**: the composition of narrowband simulation, codec quantisation, and 10 dB additive noise, representing the most realistic single condition.

These degradations serve two roles. As fixed transformations of the development set, they define the evaluation

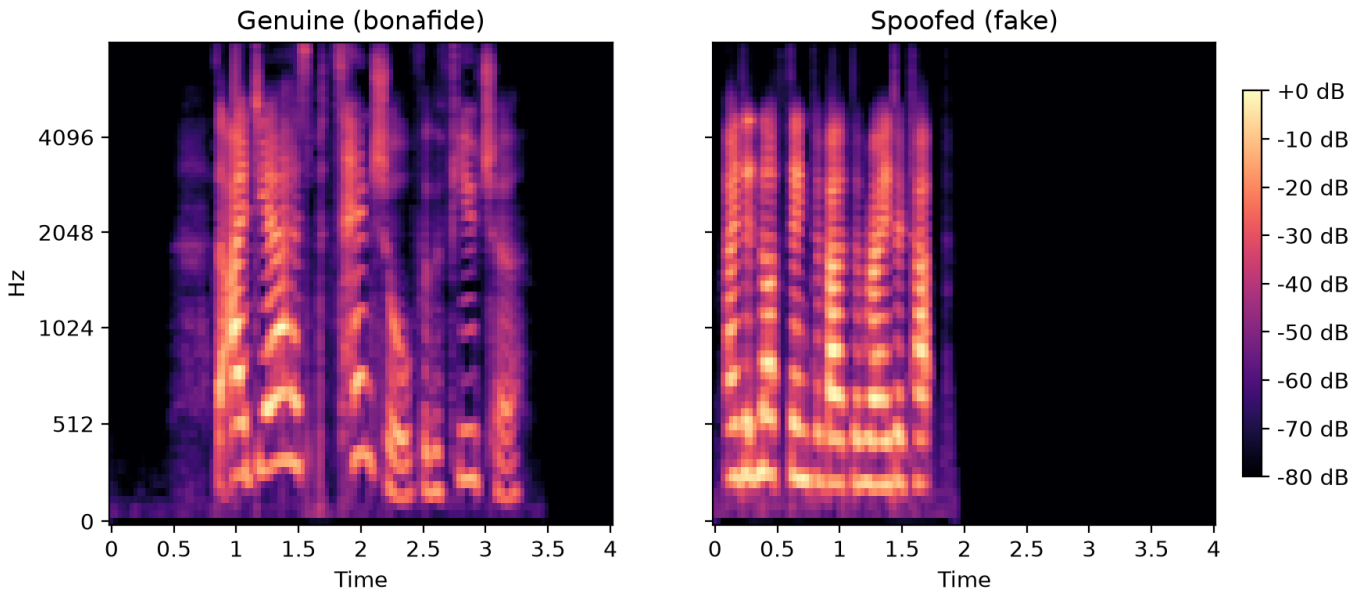


Fig. 1. Log-mel spectrograms of a genuine (left) and spoofed (right) utterance from ASVspoof 2019 LA. Synthetic speech often exhibits subtly more regular harmonic structure, though such cues are not reliably visible to the human eye—motivating a learned detector.

conditions under which robustness is measured. As a stochastic augmenter applied during training only—with probability 0.5 per clip, one degradation selected uniformly at random—they constitute the augmentation strategy under test.

D. Models

Two classifiers are compared. The baseline, MelCNN, comprises the log-mel front-end followed by three convolutional blocks (convolution, batch normalisation, ReLU, max-pooling), global average pooling, and a linear classification head, totalling approximately 23,700 trainable parameters. The high-capacity comparator is the Audio Spectrogram Transformer (AST) [2], initialised from the publicly available checkpoint pretrained on AudioSet (MIT/ast-finetuned-audioset-10-10-0.4593, 86 M parameters); its 527-way classification head is replaced with a two-class head and the full model fine-tuned. Inputs to AST are produced by its accompanying feature extractor (128 mel bands, 1,024 frames).

E. Training and evaluation

The CNN is trained with Adam at a learning rate of $1e-3$ for 5 epochs; AST is fine-tuned with AdamW at $1e-5$ for 3 epochs. In both cases, class-weighted cross-entropy compensates for the 9:1 class imbalance, and the checkpoint with the lowest development-set EER is retained. Equal Error Rate (EER) is the primary metric throughout. The degradation evaluation is computed on a fixed, randomly selected subset of 2,000 development utterances, identical across all conditions and models, so that differences in EER reflect the degradation and the model rather than the sample. Experiments were implemented in PyTorch. The CNN was trained on an Apple-Silicon (MPS) GPU, while the AST models were fine-tuned on an NVIDIA T4 GPU via Google Colab. Audio processing

used librosa and torchaudio, and the AST model and its feature extractor were obtained from the Hugging Face Transformers library. Owing to GPU quota limits on the free tier of Google Colab, the AST models were trained on a subset of the training partition rather than the full 25,380 utterances. The CNN was trained on the full training partition, whereas the AST models used a 3,000-utterance subset. The results reported for AST should therefore be regarded as preliminary, development-set figures; full-dataset training and evaluation on the held-out evaluation partition (unseen attacks A07–A19) are planned as immediate next steps.

IV. RESULTS

Table I reports the Equal Error Rate (EER) under each of the six acoustic conditions for all four systems—the CNN and AST classifiers, each in a clean-trained and a degradation-augmented variant. Figure 3 presents the same results graphically.

TABLE I
EQUAL ERROR RATE (EER, %) UNDER EACH ACOUSTIC CONDITION FOR THE CNN AND AST MODELS, COMPARING CLEAN-TRAINED AND DEGRADATION-AUGMENTED VARIANTS ON THE DEVELOPMENT SET (LOWER IS BETTER).

Condition	CNN base	CNN aug	AST base	AST aug
Clean	3.50	11.10	2.48	1.33
μ -law codec	7.90	13.41	2.15	2.58
Telephony 8 kHz	12.59	14.15	3.42	2.26
Noise (15 dB)	50.99	21.59	19.60	7.47
Noise (5 dB)	49.31	26.15	28.40	10.70
Phone call	49.77	25.74	22.04	8.58

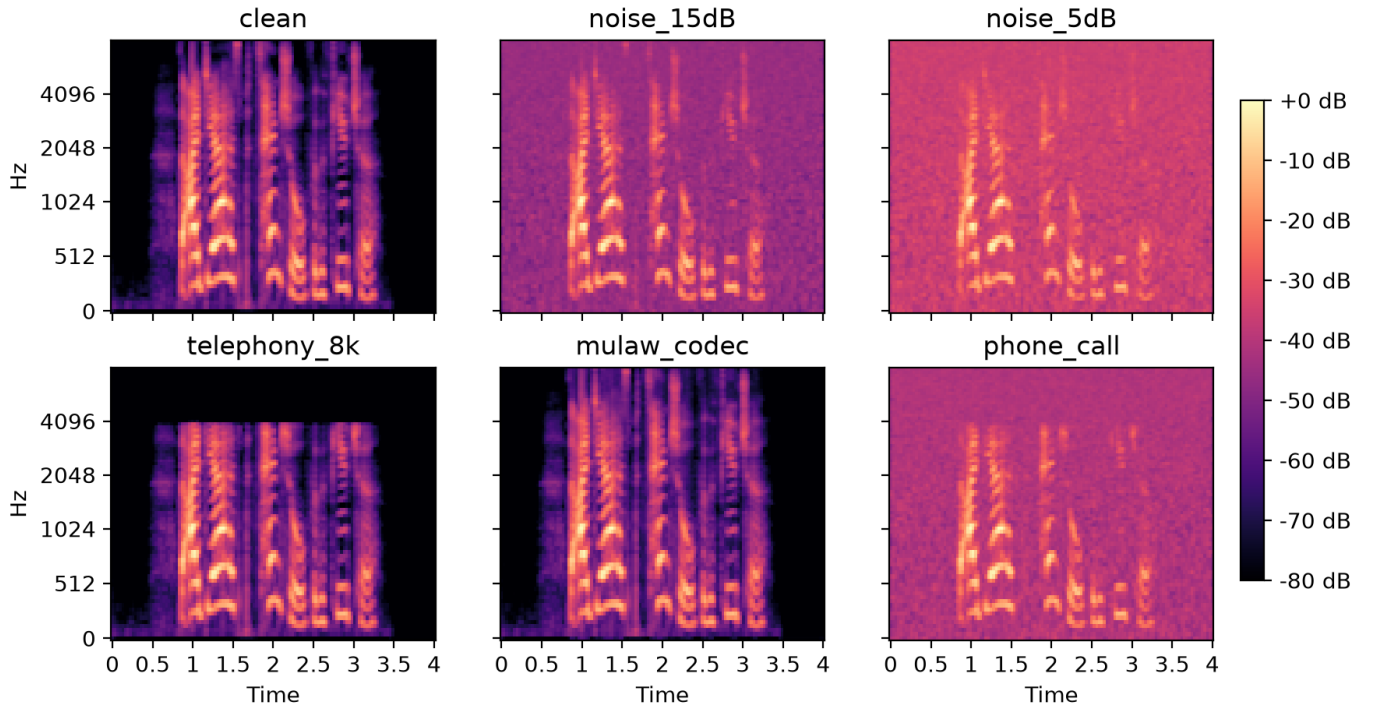


Fig. 2. The same genuine utterance under each telephony degradation. High-frequency detail is progressively lost under the narrowband and codec conditions, while additive noise fills the spectrogram, obscuring the cues a clean-trained detector relies upon.

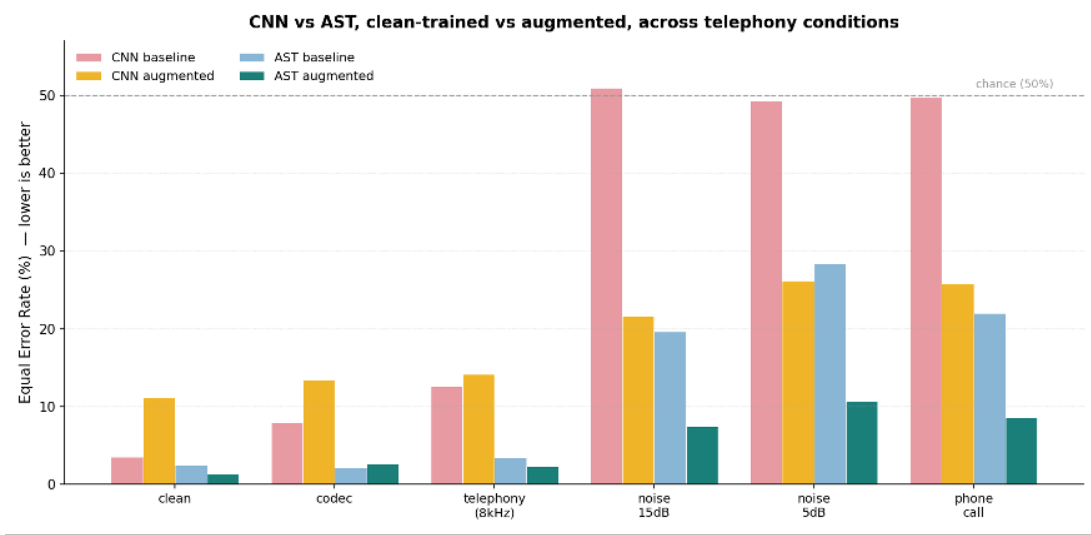


Fig. 3. EER across telephony conditions for all four models. The clean-trained CNN reaches chance under noise/phone conditions; AST with augmentation is lowest throughout

A. Clean-trained CNN baseline

On clean audio, the convolutional baseline achieves an EER of 3.50%, confirming that it learns the discrimination task competently under matched conditions. Its performance collapses, however, as soon as the audio is degraded. Under the combined phone-call condition the EER rises to 49.77%, and under additive noise it reaches 50.99% at 15 dB and 49.31% at 5 dB — values at or around the 50% chance level,

indicating a complete loss of discriminative ability. The codec and narrowband conditions are less severe (7.90% and 12.59% respectively), leaving the baseline degraded but still functional.

B. Effect of augmentation on the CNN

Training the CNN with degradation augmentation substantially recovers the collapsed conditions: the phone-call EER falls from 49.77% to 25.74%, and the noise conditions improve

from near-chance to 21.59% (15 dB) and 26.15% (5 dB). This recovery is accompanied by a cost on undistorted audio, where the EER rises from 3.50% to 11.10%; the mild codec and narrowband conditions likewise worsen slightly. The augmented CNN is thus more uniform across conditions but no longer specialised to clean audio.

C. AST Models

The clean-trained AST — fine-tuned on the reduced 3,000-utterance subset described in Section 3.E — is markedly more robust than the clean-trained CNN even without any augmentation: it records 22.04% on the phone-call condition and 19.60% at 15 dB noise, compared with near-chance values for the CNN, while matching or improving on the CNN for clean, codec, and narrowband audio. Adding augmentation yields the strongest system overall, with the lowest EER in every condition. The phone-call EER falls to 8.58% — an 83% relative reduction from the 49.77% of the clean-trained CNN baseline — and the noise conditions drop to 7.47% and 10.70%. Notably, and in contrast to the CNN, augmentation does not degrade clean-audio performance for AST but improves it, from 2.48% to 1.33%.

D. Summary of Patterns

Three patterns are evident across Table 1 and Figure 3. First, degradation drives the clean-trained CNN to chance under noise and phone-channel conditions, while affecting codec and narrowband audio only moderately. Second, augmentation restores the CNN under the severe conditions but raises its clean-audio EER. Third, the pretrained AST is more robust than the CNN throughout, and its augmented variant achieves both the lowest error under degradation and an improvement — rather than a deterioration — on clean audio. These patterns are interpreted in Section 5.

V. DISCUSSION

A. Why additive noise is catastrophic for the clean-trained CNN

The pattern in Table 1 — near-total collapse under noise, but only moderate degradation under codec and narrowband conditions — is explained by what the clean-trained model has learned to attend to. Trained exclusively on studio-quality audio, the CNN can exploit the most convenient discriminative cues available: subtle high-frequency synthesis artefacts, and the clean near-silent regions that characterise genuine recordings. Additive noise attacks precisely these cues. It floods the spectrogram with broadband energy, masking fine spectral detail and filling the silences, so that the features the model depends on are simply no longer present at test time — a textbook case of distribution shift. The codec and narrowband conditions, by contrast, remove or quantise part of the signal but leave much of its structure intact, which is why the baseline degrades under them without collapsing. Notably, the model does not fail gracefully: EER moves to approximately 50%, meaning its outputs carry no usable information under the very conditions in which a vishing detector would actually operate.

B. A capacity-driven trade-off in the small model

Augmentation restores the CNN under severe conditions but raises its clean-audio EER from 3.50% to 11.10%. We interpret this as a consequence of limited capacity. With only 23,700 parameters, the network cannot simultaneously maintain a finely tuned representation of clean audio and cover the much broader input distribution induced by augmentation; it is forced to trade specialisation for coverage. The clean-trained CNN is a specialist — excellent in its training regime, useless outside it — while the augmented CNN is a generalist that is uniformly moderate everywhere. For the vishing use case this trade is arguably worthwhile, since real calls are degraded, but it remains a genuine cost rather than a free improvement.

C. Why the pretrained transformer dissolves the trade-off

AST behaves differently in two respects. First, even without augmentation it is far more robust than the CNN (22.04% versus 49.77% on the phone-call condition), which we attribute to its pretraining: AudioSet exposes the model to a vast range of noisy, real-world recording conditions, so noise-tolerant representations are already embedded in its weights before fine-tuning begins. Second, augmentation improves AST everywhere, including on clean audio (2.48% $\rightarrow$ 1.33%): with 86 million parameters, the model has the capacity to absorb the broader training distribution without sacrificing its clean-condition representation, so the specialist-versus-generalist tension that constrains the CNN never arises. This result is made more striking by the experimental handicap noted in Section 3.E — AST was fine-tuned on roughly one-eighth of the data used to train the CNN, yet outperforms it in every condition. Taken together, these findings support the central interpretation of this work: robustness to channel degradation appears to be bounded less by the augmentation recipe than by model capacity and the breadth of pretraining.

D. Limitations

Several caveats qualify these conclusions, each pointing to concrete future work. First, the degradations are simulated: mathematically constructed noise, resampling, and companding approximate — but do not reproduce — the behaviour of real telephone networks, real codecs (e.g. AMR, Opus), or real acoustic environments, and validation on genuinely transmitted audio remains necessary. Second, the same family of degradations is used for both augmentation and evaluation, so the results demonstrate robustness to known channel conditions; robustness to unseen degradations would require holding conditions out of training, which we did not do here. Third, as discussed, the AST models were trained on a 3,000-utterance subset owing to compute constraints — although, since this biases the comparison against AST, the capacity conclusion is if anything conservative. Fourth, all results are development-set figures from single training runs: the evaluation partition, whose attacks (A07–A19) are unseen during training, has not yet been scored, and no variance across random seeds is reported. Confirming that the observed robustness transfers

to unseen attack types is the most important outstanding experiment.

VI. CONCLUSION AND FUTURE WORK

This work examined how well deepfake-audio detectors withstand the degradations of a real telephone channel — the setting in which voice-phishing detection actually matters, yet the one in which detectors are least tested. Using a controlled set of telephony degradations over ASVspoof 2019 LA, we found that a clean-trained convolutional detector collapses to near-chance under noise and combined phone-channel conditions, and that training-time augmentation with those degradations restores usable performance, albeit at a cost to clean-audio accuracy for a low-capacity model. A pretrained Audio Spectrogram Transformer proved both substantially more robust and free of that trade-off: augmentation improved its performance across every condition, clean audio included, and it did so despite being fine-tuned on a fraction of the training data. Overall, the evidence suggests that robustness to channel degradation is governed less by the augmentation recipe than by model capacity and the breadth of pretraining. Several directions follow directly from these findings. The most immediate is to complete full-dataset training and to score the held-out evaluation partition, whose attack types (A07–A19) are unseen during training, in order to confirm that the observed robustness generalises beyond the development set. Beyond the ASVspoof benchmark, evaluation on the In-the-Wild dataset of genuine and cloned speech would test the detectors against real, rather than simulated, deepfakes. The degradation model itself could be made more realistic through established augmentation schemes such as RawBoost and through real codecs (e.g. AMR and Opus) in place of the μ -law approximation used here. Finally, two controlled studies would sharpen the analysis: an ablation over the augmentation probability, tracing the robustness-versus-clean-accuracy trade-off as a curve rather than a single operating point; and a protocol that holds certain degradations out of training entirely, so that robustness to genuinely unseen channel effects — not merely to known ones — can be measured. Together these steps would move the present, preliminary findings toward a robust and deployable voice-phishing detector.

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